DBMS LAB 4

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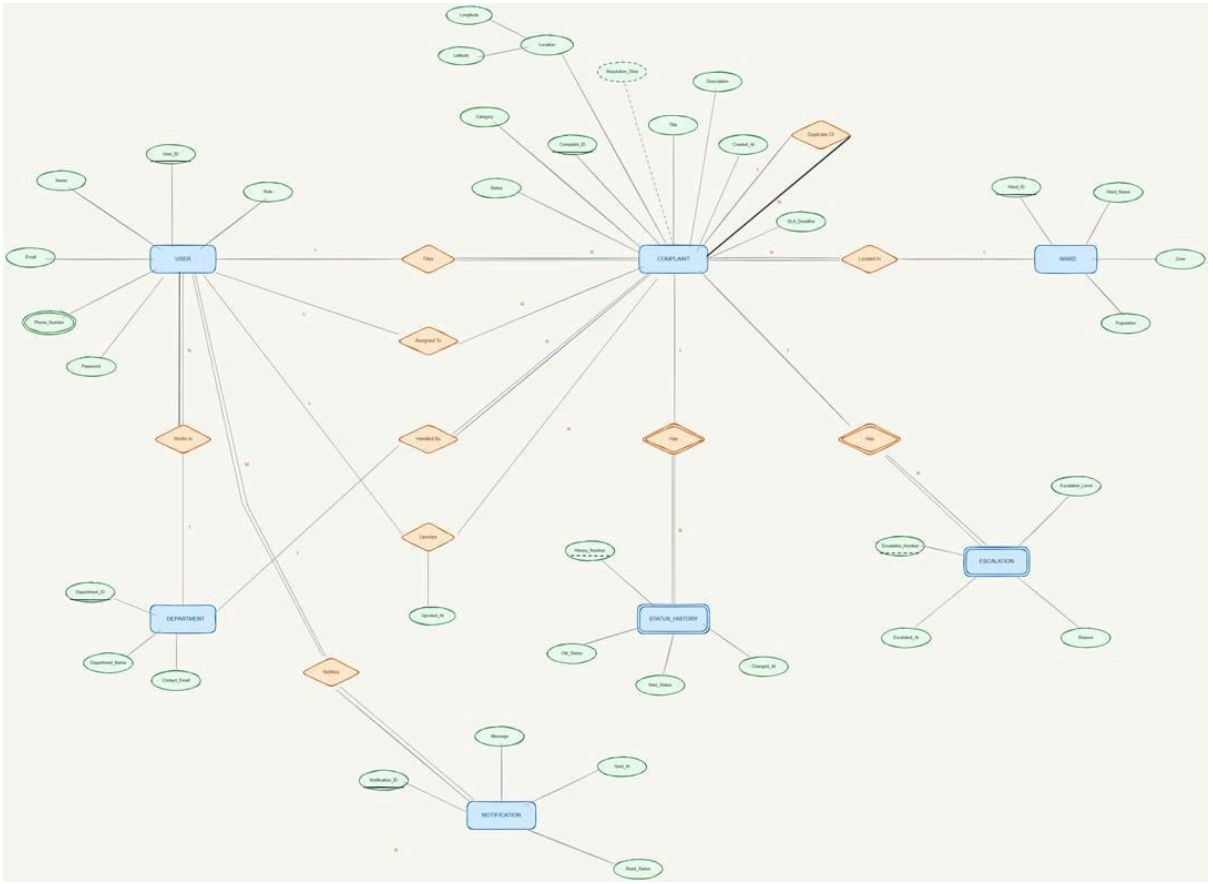
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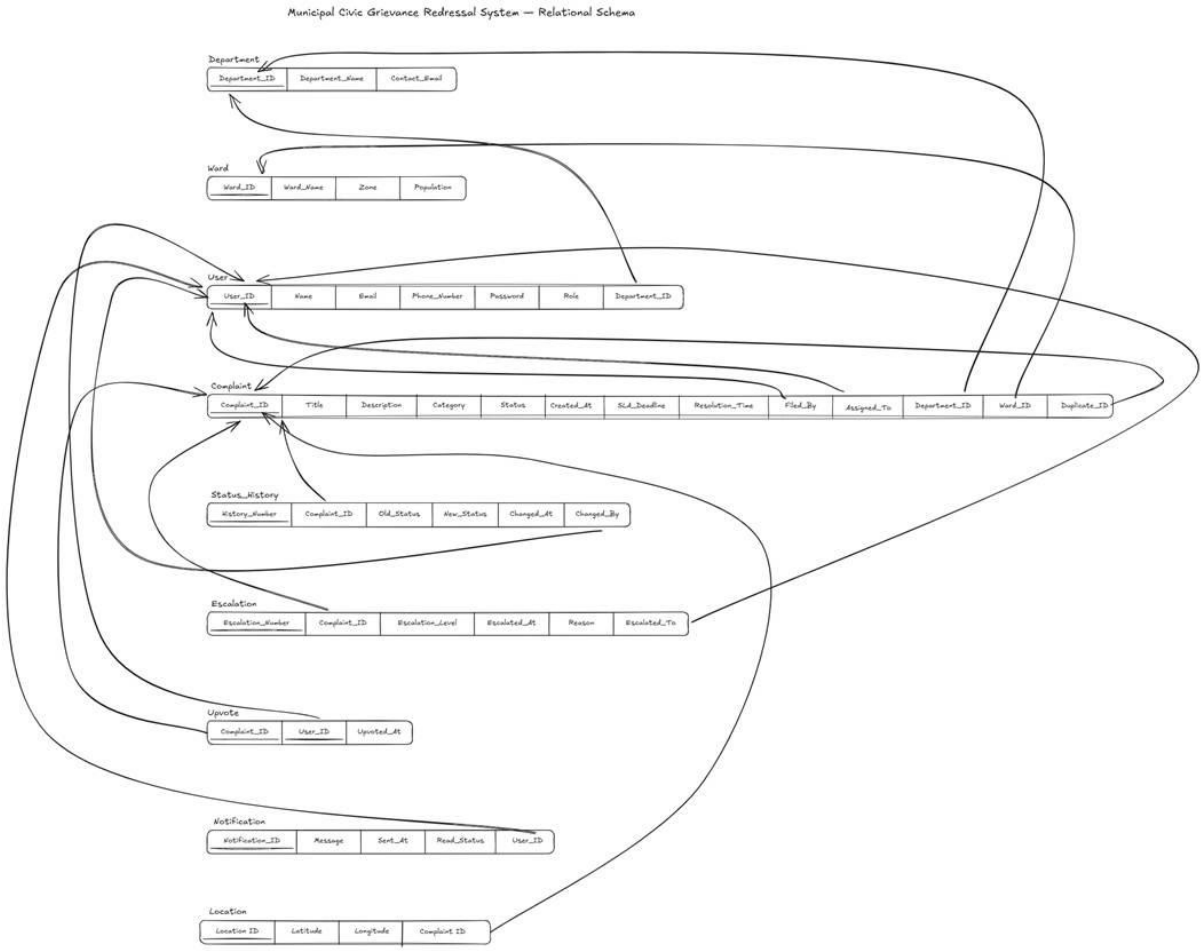
Project Title:

Municipal Civic Grievance Redressal System with Semantic Duplicate Detection and SLA-Based Auto-Escalation

ER Diagram:



Relation Schema



DDL Commands :

```
postgres=# \c civic_grievance
You are now connected to database "civic_grievance" as user "postgres".
civic_grievance=# CREATE TABLE Department (
    Department_ID   SERIAL         PRIMARY KEY,
    Department_Name VARCHAR(100)   NOT NULL,
    Contact_Email   VARCHAR(150)   NOT NULL,
    CONSTRAINT uq_department_name UNIQUE (Department_Name),
    CONSTRAINT ck_department_email CHECK (Contact_Email LIKE '%@%.%')
);

\d Department
CREATE TABLE
                                        Table "public.department"
     Column      |          Type          | Collation | Nullable |                      Default
-----------------+------------------------+-----------+----------+-----------------------------------------------
 department_id   | integer                |           | not null | nextval('department_department_id_seq'::regclass)
 department_name | character varying(100) |           | not null |
 contact_email   | character varying(150) |           | not null |
Indexes:
    "department_pkey" PRIMARY KEY, btree (department_id)
    "uq_department_name" UNIQUE CONSTRAINT, btree (department_name)
Check constraints:
    "ck_department_email" CHECK (contact_email::text ~~ '%@%.%'::text)

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE Ward (
    Ward_ID     SERIAL       PRIMARY KEY,
    Ward_Name   VARCHAR(100) NOT NULL,
    Zone        VARCHAR(50)  NOT NULL,
    Population  INTEGER      CHECK (Population >= 0),
    CONSTRAINT uq_ward_name UNIQUE (Ward_Name)
);

\d Ward
CREATE TABLE
                                      Table "public.ward"
   Column   |          Type          | Collation | Nullable |                Default
------------+------------------------+-----------+----------+---------------------------------------
 ward_id    | integer                |           | not null | nextval('ward_ward_id_seq'::regclass)
 ward_name  | character varying(100) |           | not null |
 zone       | character varying(50)  |           | not null |
 population | integer                |           |          |
Indexes:
    "ward_pkey" PRIMARY KEY, btree (ward_id)
    "uq_ward_name" UNIQUE CONSTRAINT, btree (ward_name)
Check constraints:
    "ward_population_check" CHECK (population >= 0)

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE "User" (
    User_ID       SERIAL       PRIMARY KEY,
    Name          VARCHAR(100)  NOT NULL,
    Email         VARCHAR(150)  NOT NULL,
    Phone_Number  VARCHAR(15),
    Password      VARCHAR(255)  NOT NULL,
    Role          VARCHAR(20)   NOT NULL DEFAULT 'CITIZEN',
    Department_ID INTEGER,
    CONSTRAINT uq_user_email UNIQUE (Email),
    CONSTRAINT ck_user_role     CHECK (Role IN ('CITIZEN','STAFF','ADMIN')),
    CONSTRAINT ck_user_phone_len CHECK (Phone_Number IS NULL OR LENGTH(Phone_Number) BETWEEN 7 AND 15),
    CONSTRAINT fk_user_department
        FOREIGN KEY (Department_ID) REFERENCES Department (Department_ID)
        ON UPDATE CASCADE ON DELETE SET NULL
);

\d "User"
CREATE TABLE
                                       Table "public.User"
    Column     |          Type          | Collation | Nullable |                 Default
---------------+------------------------+-----------+----------+-----------------------------------
 user_id       | integer                |           | not null | nextval('"User_user_id_seq"'::regclass)
 name          | character varying(100) |           | not null |
 email         | character varying(150) |           | not null |
 phone_number  | character varying(15)  |           |          |
 password      | character varying(255) |           | not null |
 role          | character varying(20)  |           | not null | 'CITIZEN'::character varying
 department_id | integer                |           |          |
Indexes:
    "User_pkey" PRIMARY KEY, btree (user_id)
    "uq_user_email" UNIQUE CONSTRAINT, btree (email)
Check constraints:
    "ck_user_phone_len" CHECK (phone_number IS NULL OR length(phone_number::text) >= 7 AND length(phone_number::text) <= 15)
    "ck_user_role" CHECK (role::text = ANY (ARRAY['CITIZEN'::character varying, 'STAFF'::character varying, 'ADMIN'::character varying]::text[]))
Foreign-key constraints:
    "fk_user_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE Complaint (
    Complaint_ID    SERIAL        PRIMARY KEY,
    Title           VARCHAR(150)  NOT NULL,
    Description     TEXT          NOT NULL,
    Category        VARCHAR(50)   NOT NULL,
    Status          VARCHAR(20)   NOT NULL DEFAULT 'OPEN',
    Latitude        DECIMAL(9,6),
    Longitude       DECIMAL(9,6),
    Created_At      TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    SLA_Deadline    TIMESTAMP     NOT NULL,
    Resolution_Time TIMESTAMP,
    Filed_By        INTEGER       NOT NULL,
    Assigned_To     INTEGER,
    Department_ID   INTEGER,
    Ward_ID         INTEGER       NOT NULL,
    Duplicate_Of    INTEGER,
    CONSTRAINT ck_complaint_status
        CHECK (Status IN ('OPEN','IN_PROGRESS','ESCALATED','RESOLVED','CLOSED','DUPLICATE')),
    CONSTRAINT ck_complaint_sla
        CHECK (SLA_Deadline >= Created_At),
    CONSTRAINT ck_complaint_resolution
        CHECK (Resolution_Time IS NULL OR Resolution_Time >= Created_At),
    CONSTRAINT ck_complaint_not_own_duplicate
        CHECK (Duplicate_Of IS NULL OR Duplicate_Of <> Complaint_ID),
    CONSTRAINT fk_complaint_filed_by
        FOREIGN KEY (Filed_By) REFERENCES "User" (User_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_complaint_assigned_to
        FOREIGN KEY (Assigned_To) REFERENCES "User" (User_ID)
        ON UPDATE CASCADE ON DELETE SET NULL,
    CONSTRAINT fk_complaint_department
        FOREIGN KEY (Department_ID) REFERENCES Department (Department_ID)
        ON UPDATE CASCADE ON DELETE SET NULL,
    CONSTRAINT fk_complaint_ward
        FOREIGN KEY (Ward_ID) REFERENCES Ward (Ward_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_complaint_duplicate_of
        FOREIGN KEY (Duplicate_Of) REFERENCES Complaint (Complaint_ID)
        ON UPDATE CASCADE ON DELETE SET NULL
);
```

```
\d Complaint
CREATE TABLE
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
Check constraints:
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varyin
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELET

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE Status_History (
    History_Number SERIAL       PRIMARY KEY,
    Complaint_ID   INTEGER      NOT NULL,
    Old_Status     VARCHAR(20),
    New_Status     VARCHAR(20)  NOT NULL,
    Changed_At     TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    Changed_By     INTEGER      NOT NULL,
    CONSTRAINT ck_history_status
        CHECK (New_Status IN ('OPEN','IN_PROGRESS','ESCALATED','RESOLVED','CLOSED','DUPLICATE')),
    CONSTRAINT ck_history_status_change
        CHECK (Old_Status IS NULL OR Old_Status <> New_Status),
    CONSTRAINT fk_history_complaint
        FOREIGN KEY (Complaint_ID) REFERENCES Complaint (Complaint_ID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_history_changed_by
        FOREIGN KEY (Changed_By) REFERENCES "User" (User_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);

\d Status_History
CREATE TABLE
                                          Table "public.status_history"
     Column     |            Type             | Collation | Nullable |                      Default
----------------+-----------------------------+-----------+----------+-----------------------------------------------
 history_number | integer                     |           | not null | nextval('status_history_history_number_seq'::regclass)
 complaint_id   | integer                     |           | not null |
 old_status     | character varying(20)       |           |          |
 new_status     | character varying(20)       |           | not null |
 changed_at     | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 changed_by     | integer                     |           | not null |
Indexes:
    "status_history_pkey" PRIMARY KEY, btree (history_number)
Check constraints:
    "ck_history_status" CHECK (new_status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying,
 'CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
    "ck_history_status_change" CHECK (old_status IS NULL OR old_status::text <> new_status::text)
Foreign-key constraints:
    "fk_history_changed_by" FOREIGN KEY (changed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE Escalation (
    Escalation_Number SERIAL       PRIMARY KEY,
    Complaint_ID       INTEGER      NOT NULL,
    Escalation_Level   INTEGER      NOT NULL DEFAULT 1,
    Escalated_At       TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    Reason             VARCHAR(255) NOT NULL,
    Escalated_To       INTEGER      NOT NULL,
    CONSTRAINT ck_escalation_level CHECK (Escalation_Level BETWEEN 1 AND 5),
    CONSTRAINT fk_escalation_complaint
        FOREIGN KEY (Complaint_ID) REFERENCES Complaint (Complaint_ID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_escalation_escalated_to
        FOREIGN KEY (Escalated_To) REFERENCES "User" (User_ID)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT uq_escalation_complaint_level UNIQUE (Complaint_ID, Escalation_Level)
);

\d Escalation
CREATE TABLE
                                            Table "public.escalation"
      Column       |            Type             | Collation | Nullable |                        Default
-------------------+-----------------------------+-----------+----------+-------------------------------------------------------
 escalation_number | integer                     |           | not null | nextval('escalation_escalation_number_seq'::regclass)
 complaint_id      | integer                     |           | not null |
 escalation_level  | integer                     |           | not null | 1
 escalated_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 reason            | character varying(255)      |           | not null |
 escalated_to      | integer                     |           | not null |
Indexes:
    "escalation_pkey" PRIMARY KEY, btree (escalation_number)
    "uq_escalation_complaint_level" UNIQUE CONSTRAINT, btree (complaint_id, escalation_level)
Check constraints:
    "ck_escalation_level" CHECK (escalation_level >= 1 AND escalation_level <= 5)
Foreign-key constraints:
    "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_escalation_escalated_to" FOREIGN KEY (escalated_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE Upvote (
    Complaint_ID INTEGER   NOT NULL,
    User_ID      INTEGER   NOT NULL,
    Upvoted_At   TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_upvote PRIMARY KEY (Complaint_ID, User_ID),
    CONSTRAINT fk_upvote_complaint
        FOREIGN KEY (Complaint_ID) REFERENCES Complaint (Complaint_ID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_upvote_user
        FOREIGN KEY (User_ID) REFERENCES "User" (User_ID)
        ON UPDATE CASCADE ON DELETE CASCADE
);

\d Upvote
CREATE TABLE
                              Table "public.upvote"
    Column    |            Type             | Collation | Nullable |      Default
--------------+-----------------------------+-----------+----------+-------------------
 complaint_id | integer                     |           | not null |
 user_id      | integer                     |           | not null |
 upvoted_at   | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
Indexes:
    "pk_upvote" PRIMARY KEY, btree (complaint_id, user_id)
Foreign-key constraints:
    "fk_upvote_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_upvote_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# CREATE TABLE Notification (
    Notification_ID SERIAL       PRIMARY KEY,
    Message         VARCHAR(255) NOT NULL,
    Sent_At         TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    Read_Status     BOOLEAN      NOT NULL DEFAULT FALSE,
    User_ID         INTEGER      NOT NULL,
    Department_ID   INTEGER,
    CONSTRAINT fk_notification_user
        FOREIGN KEY (User_ID) REFERENCES "User" (User_ID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_notification_department
        FOREIGN KEY (Department_ID) REFERENCES Department (Department_ID)
        ON UPDATE CASCADE ON DELETE SET NULL
);

\d Notification
CREATE TABLE
                                          Table "public.notification"
     Column      |            Type             | Collation | Nullable |                         Default
-----------------+-----------------------------+-----------+----------+---------------------------------------------------------
 notification_id | integer                     |           | not null | nextval('notification_notification_id_seq'::regclass)
 message         | character varying(255)      |           | not null |
 sent_at         | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 read_status     | boolean                     |           | not null | false
 user_id         | integer                     |           | not null |
 department_id   | integer                     |           |          |
Indexes:
    "notification_pkey" PRIMARY KEY, btree (notification_id)
Foreign-key constraints:
    "fk_notification_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_notification_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# ALTER TABLE Complaint
    ADD COLUMN Duplicate_Score DECIMAL(4,3);

\d Complaint
ALTER TABLE
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                     Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
Check constraints:
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "status_history" CONSTRAINT "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "upvote" CONSTRAINT "fk_upvote_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=# ALTER TABLE Complaint
    ADD CONSTRAINT ck_complaint_duplicate_score
    CHECK (Duplicate_Score IS NULL OR Duplicate_Score BETWEEN 0 AND 1);

\d Complaint
ALTER TABLE
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                     Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=# ALTER TABLE Complaint
    ADD COLUMN Assigned_At TIMESTAMP;

\d Complaint
ALTER TABLE
                                          Table "public.complaint"
     Column      |           Type             | Collation | Nullable |                    Default
-----------------+----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                    |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)     |           | not null |
 description     | text                       |           | not null |
 category        | character varying(50)      |           | not null |
 status          | character varying(20)      |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)               |           |          |
 longitude       | numeric(9,6)               |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                    |           | not null |
 assigned_to     | integer                    |           |          |
 department_id   | integer                    |           |          |
 ward_id         | integer                    |           | not null |
 duplicate_of    | integer                    |           |          |
 duplicate_score | numeric(4,3)               |           |          |
 assigned_at     | timestamp without time zone |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "status_history" CONSTRAINT "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=# ALTER TABLE Escalation
    ALTER COLUMN Reason TYPE VARCHAR(500);

\d Escalation
ALTER TABLE
                                            Table "public.escalation"
      Column       |            Type             | Collation | Nullable |                       Default
-------------------+-----------------------------+-----------+----------+-----------------------------------------------------
 escalation_number | integer                     |           | not null | nextval('escalation_escalation_number_seq'::regclass)
 complaint_id      | integer                     |           | not null |
 escalation_level  | integer                     |           | not null | 1
 escalated_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 reason            | character varying(500)      |           | not null |
 escalated_to      | integer                     |           | not null |
Indexes:
    "escalation_pkey" PRIMARY KEY, btree (escalation_number)
    "uq_escalation_complaint_level" UNIQUE CONSTRAINT, btree (complaint_id, escalation_level)
Check constraints:
    "ck_escalation_level" CHECK (escalation_level >= 1 AND escalation_level <= 5)
Foreign-key constraints:
    "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_escalation_escalated_to" FOREIGN KEY (escalated_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT

civic_grievance=#
```

```
civic_grievance=# ALTER TABLE Ward
    RENAME COLUMN Population TO Ward_Population;

\d Ward
ALTER TABLE
                                       Table "public.ward"
     Column      |          Type          | Collation | Nullable |                Default
-----------------+------------------------+-----------+----------+---------------------------------------
 ward_id         | integer                |           | not null | nextval('ward_ward_id_seq'::regclass)
 ward_name       | character varying(100) |           | not null |
 zone            | character varying(50)  |           | not null |
 ward_population | integer                |           |          |
Indexes:
    "ward_pkey" PRIMARY KEY, btree (ward_id)
    "uq_ward_name" UNIQUE CONSTRAINT, btree (ward_name)
Check constraints:
    "ward_population_check" CHECK (ward_population >= 0)
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT

civic_grievance=#
```

```
civic_grievance=# ALTER TABLE Notification
    ALTER COLUMN Read_Status SET DEFAULT FALSE;

\d Notification
ALTER TABLE
                                          Table "public.notification"
     Column      |            Type             | Collation | Nullable |                       Default
-----------------+-----------------------------+-----------+----------+-------------------------------------------------
 notification_id | integer                     |           | not null | nextval('notification_notification_id_seq'::regclass)
 message         | character varying(255)      |           | not null |
 sent_at         | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 read_status     | boolean                     |           | not null | false
 user_id         | integer                     |           | not null |
 department_id   | integer                     |           |          |
Indexes:
    "notification_pkey" PRIMARY KEY, btree (notification_id)
Foreign-key constraints:
    "fk_notification_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_notification_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# ALTER TABLE "User"
    ADD CONSTRAINT ck_staff_has_department
    CHECK (Role = 'CITIZEN' OR Department_ID IS NOT NULL);

\d "User"
ALTER TABLE
                                      Table "public.User"
    Column     |          Type          | Collation | Nullable |                Default
---------------+------------------------+-----------+----------+---------------------------------------
 user_id       | integer                |           | not null | nextval('"User_user_id_seq"'::regclass)
 name          | character varying(100) |           | not null |
 email         | character varying(150) |           | not null |
 phone_number  | character varying(15)  |           |          |
 password      | character varying(255) |           | not null |
 role          | character varying(20)  |           | not null | 'CITIZEN'::character varying
 department_id | integer                |           |          |
Indexes:
    "User_pkey" PRIMARY KEY, btree (user_id)
    "uq_user_email" UNIQUE CONSTRAINT, btree (email)
Check constraints:
    "ck_staff_has_department" CHECK (role::text = 'CITIZEN'::text OR department_id IS NOT NULL)
    "ck_user_phone_len" CHECK (phone_number IS NULL OR length(phone_number::text) >= 7 AND length(phone_number::text) <= 15)
    "ck_user_role" CHECK (role::text = ANY (ARRAY['CITIZEN'::character varying, 'STAFF'::character varying, 'ADMIN'::character varying]::text[]))
Foreign-key constraints:
    "fk_user_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "complaint" CONSTRAINT "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    TABLE "escalation" CONSTRAINT "fk_escalation_escalated_to" FOREIGN KEY (escalated_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    TABLE "status_history" CONSTRAINT "fk_history_changed_by" FOREIGN KEY (changed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    TABLE "notification" CONSTRAINT "fk_notification_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "upvote" CONSTRAINT "fk_upvote_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# ALTER TABLE Complaint
    DROP COLUMN Assigned_At;

\d Complaint
ALTER TABLE
                                       Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "status_history" CONSTRAINT "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "upvote" CONSTRAINT "fk_upvote_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=#
CREATE INDEX idx_user_department ON "User" (Department_ID);

\d "User"
CREATE INDEX
                                        Table "public.User"
    Column     |          Type          | Collation | Nullable |                  Default
---------------+------------------------+-----------+----------+-----------------------------------------
 user_id       | integer                |           | not null | nextval('"User_user_id_seq"'::regclass)
 name          | character varying(100) |           | not null |
 email         | character varying(150) |           | not null |
 phone_number  | character varying(15)  |           |          |
 password      | character varying(255) |           | not null |
 role          | character varying(20)  |           | not null | 'CITIZEN'::character varying
 department_id | integer                |           |          |
Indexes:
    "User_pkey" PRIMARY KEY, btree (user_id)
    "idx_user_department" btree (department_id)
    "uq_user_email" UNIQUE CONSTRAINT, btree (email)
Check constraints:
    "ck_staff_has_department" CHECK (role::text = 'CITIZEN'::text OR department_id IS NOT NULL)
    "ck_user_phone_len" CHECK (phone_number IS NULL OR length(phone_number::text) >= 7 AND length(phone_number::text) <= 15)
    "ck_user_role" CHECK (role::text = ANY (ARRAY['CITIZEN'::character varying, 'STAFF'::character varying, 'ADMIN'::character varying]::text[]))
Foreign-key constraints:
    "fk_user_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "complaint" CONSTRAINT "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    TABLE "escalation" CONSTRAINT "fk_escalation_escalated_to" FOREIGN KEY (escalated_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    TABLE "status_history" CONSTRAINT "fk_history_changed_by" FOREIGN KEY (changed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    TABLE "notification" CONSTRAINT "fk_notification_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "upvote" CONSTRAINT "fk_upvote_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# CREATE INDEX idx_complaint_filed_by ON Complaint (Filed_By);

\d Complaint
CREATE INDEX
                                         Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                     Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_filed_by" btree (filed_by)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "status_history" CONSTRAINT "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "upvote" CONSTRAINT "fk_upvote_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=# CREATE INDEX idx_complaint_assigned_to ON Complaint (Assigned_To);

\d Complaint
CREATE INDEX
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_filed_by" btree (filed_by)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "status_history" CONSTRAINT "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=# CREATE INDEX idx_complaint_department ON Complaint (Department_ID);

\d Complaint
CREATE INDEX
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_department" btree (department_id)
    "idx_complaint_filed_by" btree (filed_by)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    TABLE "escalation" CONSTRAINT "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "status_history" CONSTRAINT "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
```

```
civic_grievance=# CREATE INDEX idx_complaint_ward ON Complaint (Ward_ID);

\d Complaint
CREATE INDEX
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_department" btree (department_id)
    "idx_complaint_filed_by" btree (filed_by)
    "idx_complaint_ward" btree (ward_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
```

```
civic_grievance=# CREATE INDEX idx_complaint_duplicate_of ON Complaint (Duplicate_Of);

\d Complaint
CREATE INDEX
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_department" btree (department_id)
    "idx_complaint_duplicate_of" btree (duplicate_of)
    "idx_complaint_filed_by" btree (filed_by)
    "idx_complaint_ward" btree (ward_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
```

```
civic_grievance=#
CREATE INDEX idx_history_complaint ON Status_History (Complaint_ID);

\d Status_History
CREATE INDEX
                                        Table "public.status_history"
     Column     |            Type             | Collation | Nullable |                       Default
----------------+-----------------------------+-----------+----------+-------------------------------------------------------
 history_number | integer                     |           | not null | nextval('status_history_history_number_seq'::regclass)
 complaint_id   | integer                     |           | not null |
 old_status     | character varying(20)       |           |          |
 new_status     | character varying(20)       |           | not null |
 changed_at     | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 changed_by     | integer                     |           | not null |
Indexes:
    "status_history_pkey" PRIMARY KEY, btree (history_number)
    "idx_history_complaint" btree (complaint_id)
Check constraints:
    "ck_history_status" CHECK (new_status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying,
 'CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
    "ck_history_status_change" CHECK (old_status IS NULL OR old_status::text <> new_status::text)
Foreign-key constraints:
    "fk_history_changed_by" FOREIGN KEY (changed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# CREATE INDEX idx_history_changed_by ON Status_History (Changed_By);

\d Status_History
CREATE INDEX
                                          Table "public.status_history"
     Column     |            Type             | Collation | Nullable |                       Default
----------------+-----------------------------+-----------+----------+-------------------------------------------------------
 history_number | integer                     |           | not null | nextval('status_history_history_number_seq'::regclass)
 complaint_id   | integer                     |           | not null |
 old_status     | character varying(20)       |           |          |
 new_status     | character varying(20)       |           | not null |
 changed_at     | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 changed_by     | integer                     |           | not null |
Indexes:
    "status_history_pkey" PRIMARY KEY, btree (history_number)
    "idx_history_changed_by" btree (changed_by)
    "idx_history_complaint" btree (complaint_id)
Check constraints:
    "ck_history_status" CHECK (new_status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying,
 'CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
    "ck_history_status_change" CHECK (old_status IS NULL OR old_status::text <> new_status::text)
Foreign-key constraints:
    "fk_history_changed_by" FOREIGN KEY (changed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
    "fk_history_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=# CREATE INDEX idx_escalation_complaint ON Escalation (Complaint_ID);

\d Escalation
CREATE INDEX
                                            Table "public.escalation"
      Column       |            Type             | Collation | Nullable |                        Default
-------------------+-----------------------------+-----------+----------+-------------------------------------------------------
 escalation_number | integer                     |           | not null | nextval('escalation_escalation_number_seq'::regclass)
 complaint_id      | integer                     |           | not null |
 escalation_level  | integer                     |           | not null | 1
 escalated_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 reason            | character varying(500)      |           | not null |
 escalated_to      | integer                     |           | not null |
Indexes:
    "escalation_pkey" PRIMARY KEY, btree (escalation_number)
    "idx_escalation_complaint" btree (complaint_id)
    "uq_escalation_complaint_level" UNIQUE CONSTRAINT, btree (complaint_id, escalation_level)
Check constraints:
    "ck_escalation_level" CHECK (escalation_level >= 1 AND escalation_level <= 5)
Foreign-key constraints:
    "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_escalation_escalated_to" FOREIGN KEY (escalated_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT

civic_grievance=#
```

```
civic_grievance=# CREATE INDEX idx_escalation_escalated_to ON Escalation (Escalated_To);

\d Escalation
CREATE INDEX
                                            Table "public.escalation"
      Column       |            Type             | Collation | Nullable |                        Default
-------------------+-----------------------------+-----------+----------+-------------------------------------------------------
 escalation_number | integer                     |           | not null | nextval('escalation_escalation_number_seq'::regclass)
 complaint_id      | integer                     |           | not null |
 escalation_level  | integer                     |           | not null | 1
 escalated_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 reason            | character varying(500)      |           | not null |
 escalated_to      | integer                     |           | not null |
Indexes:
    "escalation_pkey" PRIMARY KEY, btree (escalation_number)
    "idx_escalation_complaint" btree (complaint_id)
    "idx_escalation_escalated_to" btree (escalated_to)
    "uq_escalation_complaint_level" UNIQUE CONSTRAINT, btree (complaint_id, escalation_level)
Check constraints:
    "ck_escalation_level" CHECK (escalation_level >= 1 AND escalation_level <= 5)
Foreign-key constraints:
    "fk_escalation_complaint" FOREIGN KEY (complaint_id) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_escalation_escalated_to" FOREIGN KEY (escalated_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT

civic_grievance=#
```

```
civic_grievance=# CREATE INDEX idx_notification_user ON Notification (User_ID);

\d Notification
CREATE INDEX
                                          Table "public.notification"
     Column      |            Type             | Collation | Nullable |                       Default
-----------------+-----------------------------+-----------+----------+-------------------------------------------------------
 notification_id | integer                     |           | not null | nextval('notification_notification_id_seq'::regclass)
 message         | character varying(255)      |           | not null |
 sent_at         | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 read_status     | boolean                     |           | not null | false
 user_id         | integer                     |           | not null |
 department_id   | integer                     |           |          |
Indexes:
    "notification_pkey" PRIMARY KEY, btree (notification_id)
    "idx_notification_user" btree (user_id)
Foreign-key constraints:
    "fk_notification_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_notification_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

```
civic_grievance=# CREATE INDEX idx_notification_department ON Notification (Department_ID);

\d Notification
CREATE INDEX
                                          Table "public.notification"
     Column      |            Type             | Collation | Nullable |                      Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------------
 notification_id | integer                     |           | not null | nextval('notification_notification_id_seq'::regclass)
 message         | character varying(255)      |           | not null |
 sent_at         | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 read_status     | boolean                     |           | not null | false
 user_id         | integer                     |           | not null |
 department_id   | integer                     |           |          |
Indexes:
    "notification_pkey" PRIMARY KEY, btree (notification_id)
    "idx_notification_department" btree (department_id)
    "idx_notification_user" btree (user_id)
Foreign-key constraints:
    "fk_notification_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_notification_user" FOREIGN KEY (user_id) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE CASCADE

civic_grievance=#
```

SQL

```sql
CREATE INDEX idx_complaint_category_ward_created ON Complaint (Category, Ward_ID, Created_At);
\d Complaint
```

```
                                          Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                     Default
-----------------+-----------------------------+-----------+----------+-------------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_category_ward_created" btree (category, ward_id, created_at)
    "idx_complaint_department" btree (department_id)
    "idx_complaint_duplicate_of" btree (duplicate_of)
    "idx_complaint_filed_by" btree (filed_by)
    "idx_complaint_sla_status" btree (status, sla_deadline)
    "idx_complaint_ward" btree (ward_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
:
```

SQL

```sql
DROP INDEX IF EXISTS idx_complaint_category_ward_created;
\d Complaint
```

```
                                        Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_department" btree (department_id)
    "idx_complaint_duplicate_of" btree (duplicate_of)
    "idx_complaint_filed_by" btree (filed_by)
    "idx_complaint_sla_status" btree (status, sla_deadline)
    "idx_complaint_ward" btree (ward_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
    TABLE "complaint" CONSTRAINT "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
:
```

SQL

```sql
CREATE INDEX idx_complaint_category_ward_created ON Complaint
(Category, Ward_ID, Created_At DESC);
\d Complaint
```

```
                                        Table "public.complaint"
     Column      |            Type             | Collation | Nullable |                    Default
-----------------+-----------------------------+-----------+----------+-----------------------------------------------
 complaint_id    | integer                     |           | not null | nextval('complaint_complaint_id_seq'::regclass)
 title           | character varying(150)      |           | not null |
 description     | text                        |           | not null |
 category        | character varying(50)       |           | not null |
 status          | character varying(20)       |           | not null | 'OPEN'::character varying
 latitude        | numeric(9,6)                |           |          |
 longitude       | numeric(9,6)                |           |          |
 created_at      | timestamp without time zone |           | not null | CURRENT_TIMESTAMP
 sla_deadline    | timestamp without time zone |           | not null |
 resolution_time | timestamp without time zone |           |          |
 filed_by        | integer                     |           | not null |
 assigned_to     | integer                     |           |          |
 department_id   | integer                     |           |          |
 ward_id         | integer                     |           | not null |
 duplicate_of    | integer                     |           |          |
 duplicate_score | numeric(4,3)                |           |          |
Indexes:
    "complaint_pkey" PRIMARY KEY, btree (complaint_id)
    "idx_complaint_assigned_to" btree (assigned_to)
    "idx_complaint_category_ward_created" btree (category, ward_id, created_at DESC)
    "idx_complaint_department" btree (department_id)
    "idx_complaint_duplicate_of" btree (duplicate_of)
    "idx_complaint_filed_by" btree (filed_by)
    "idx_complaint_sla_status" btree (status, sla_deadline)
    "idx_complaint_ward" btree (ward_id)
Check constraints:
    "ck_complaint_duplicate_score" CHECK (duplicate_score IS NULL OR duplicate_score >= 0::numeric AND duplicate_score <= 1::numeric)
    "ck_complaint_not_own_duplicate" CHECK (duplicate_of IS NULL OR duplicate_of <> complaint_id)
    "ck_complaint_resolution" CHECK (resolution_time IS NULL OR resolution_time >= created_at)
    "ck_complaint_sla" CHECK (sla_deadline >= created_at)
    "ck_complaint_status" CHECK (status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying, 'RESOLVED'::character varying, '
CLOSED'::character varying, 'DUPLICATE'::character varying]::text[]))
Foreign-key constraints:
    "fk_complaint_assigned_to" FOREIGN KEY (assigned_to) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_department" FOREIGN KEY (department_id) REFERENCES department(department_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_duplicate_of" FOREIGN KEY (duplicate_of) REFERENCES complaint(complaint_id) ON UPDATE CASCADE ON DELETE SET NULL
    "fk_complaint_filed_by" FOREIGN KEY (filed_by) REFERENCES "User"(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
    "fk_complaint_ward" FOREIGN KEY (ward_id) REFERENCES ward(ward_id) ON UPDATE CASCADE ON DELETE RESTRICT
Referenced by:
:
```

```
civic_grievance=# CREATE VIEW vw_sla_breach_candidates AS
SELECT
    c.Complaint_ID,
    c.Title,
    c.Category,
    c.Status,
    c.SLA_Deadline,
    c.Department_ID,
    d.Department_Name,
    c.Ward_ID,
    EXTRACT(EPOCH FROM (CURRENT_TIMESTAMP - c.SLA_Deadline)) / 3600 AS Hours_Overdue
FROM Complaint c
LEFT JOIN Department d ON d.Department_ID = c.Department_ID
WHERE c.Status IN ('OPEN', 'IN_PROGRESS')
  AND c.SLA_Deadline < CURRENT_TIMESTAMP;

\d+ vw_sla_breach_candidates
CREATE VIEW
                                View "public.vw_sla_breach_candidates"
     Column      |            Type             | Collation | Nullable | Default | Storage  | Description
-----------------+-----------------------------+-----------+----------+---------+----------+-------------
 complaint_id    | integer                     |           |          |         | plain    |
 title           | character varying(150)      |           |          |         | extended |
 category        | character varying(50)       |           |          |         | extended |
 status          | character varying(20)       |           |          |         | extended |
 sla_deadline    | timestamp without time zone |           |          |         | plain    |
 department_id   | integer                     |           |          |         | plain    |
 department_name | character varying(100)      |           |          |         | extended |
 ward_id         | integer                     |           |          |         | plain    |
 hours_overdue   | numeric                     |           |          |         | main     |
View definition:
 SELECT c.complaint_id,
    c.title,
    c.category,
    c.status,
    c.sla_deadline,
    c.department_id,
    d.department_name,
    c.ward_id,
    EXTRACT(epoch FROM CURRENT_TIMESTAMP - c.sla_deadline::timestamp with time zone) / 3600::numeric AS hours_overdue
   FROM complaint c
     LEFT JOIN department d ON d.department_id = c.department_id
  WHERE (c.status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying]::text[])) AND c.sla_deadline < CURRENT_TIMESTAMP;
```

```
civic_grievance=# CREATE VIEW vw_complaint_full_detail AS
SELECT
    c.Complaint_ID,
    c.Title,
    c.Category,
    c.Status,
    c.Created_At,
    c.SLA_Deadline,
    c.Resolution_Time,
    filer.Name  AS Filed_By_Name,
    staff.Name  AS Assigned_To_Name,
    d.Department_Name,
    w.Ward_Name,
    w.Zone,
    c.Duplicate_Of,
    (SELECT COUNT(*) FROM Upvote u WHERE u.Complaint_ID = c.Complaint_ID) AS Upvote_Count
FROM Complaint c
JOIN "User" filer      ON filer.User_ID = c.Filed_By
LEFT JOIN "User" staff  ON staff.User_ID = c.Assigned_To
LEFT JOIN Department d ON d.Department_ID = c.Department_ID
JOIN Ward w             ON w.Ward_ID = c.Ward_ID;

\d+ vw_complaint_full_detail
CREATE VIEW
                                View "public.vw_complaint_full_detail"
      Column      |            Type             | Collation | Nullable | Default | Storage  | Description
------------------+-----------------------------+-----------+----------+---------+----------+-------------
 complaint_id     | integer                     |           |          |         | plain    |
 title            | character varying(150)      |           |          |         | extended |
 category         | character varying(50)       |           |          |         | extended |
 status           | character varying(20)       |           |          |         | extended |
 created_at       | timestamp without time zone |           |          |         | plain    |
 sla_deadline     | timestamp without time zone |           |          |         | plain    |
 resolution_time  | timestamp without time zone |           |          |         | plain    |
 filed_by_name    | character varying(100)      |           |          |         | extended |
 assigned_to_name | character varying(100)      |           |          |         | extended |
 department_name  | character varying(100)      |           |          |         | extended |
 ward_name        | character varying(100)      |           |          |         | extended |
 zone             | character varying(50)       |           |          |         | extended |
 duplicate_of     | integer                     |           |          |         | plain    |
 upvote_count     | bigint                      |           |          |         | plain    |
View definition:
 SELECT c.complaint_id,
```

```
View definition:
 SELECT c.complaint_id,
    c.title,
    c.category,
    c.status,
    c.created_at,
    c.sla_deadline,
    c.resolution_time,
    filer.name AS filed_by_name,
    staff.name AS assigned_to_name,
    d.department_name,
    w.ward_name,
    w.zone,
    c.duplicate_of,
    ( SELECT count(*) AS count
           FROM upvote u
          WHERE u.complaint_id = c.complaint_id) AS upvote_count
   FROM complaint c
     JOIN "User" filer ON filer.user_id = c.filed_by
     LEFT JOIN "User" staff ON staff.user_id = c.assigned_to
     LEFT JOIN department d ON d.department_id = c.department_id
     JOIN ward w ON w.ward_id = c.ward_id;

civic_grievance=#
```

```
civic_grievance=# CREATE VIEW vw_escalation_chain AS
SELECT
    e.Complaint_ID,
    c.Title,
    e.Escalation_Number,
    e.Escalation_Level,
    e.Escalated_At,
    e.Reason,
    esc_to.Name AS Escalated_To_Name,
    esc_to.Role AS Escalated_To_Role
FROM Escalation e
JOIN Complaint c    ON c.Complaint_ID = e.Complaint_ID
JOIN "User" esc_to  ON esc_to.User_ID = e.Escalated_To
ORDER BY e.Complaint_ID, e.Escalation_Level;
CREATE VIEW
civic_grievance=# \d+ vw_escalation_chain
                                    View "public.vw_escalation_chain"
      Column       |            Type             | Collation | Nullable | Default | Storage  | Description
-------------------+-----------------------------+-----------+----------+---------+----------+-------------
 complaint_id      | integer                     |           |          |         | plain    |
 title             | character varying(150)      |           |          |         | extended |
 escalation_number | integer                     |           |          |         | plain    |
 escalation_level  | integer                     |           |          |         | plain    |
 escalated_at      | timestamp without time zone |           |          |         | plain    |
 reason            | character varying(500)      |           |          |         | extended |
 escalated_to_name | character varying(100)      |           |          |         | extended |
 escalated_to_role | character varying(20)       |           |          |         | extended |
View definition:
 SELECT e.complaint_id,
    c.title,
    e.escalation_number,
    e.escalation_level,
    e.escalated_at,
    e.reason,
    esc_to.name AS escalated_to_name,
    esc_to.role AS escalated_to_role
   FROM escalation e
     JOIN complaint c ON c.complaint_id = e.complaint_id
     JOIN "User" esc_to ON esc_to.user_id = e.escalated_to
  ORDER BY e.complaint_id, e.escalation_level;

civic_grievance=#
```

```
civic_grievance=# CREATE VIEW vw_department_workload AS
SELECT
    d.Department_ID,
    d.Department_Name,
    COUNT(*) FILTER (WHERE c.Status IN ('OPEN','IN_PROGRESS','ESCALATED')) AS Open_Complaints,
    COUNT(*) FILTER (WHERE c.Status IN ('RESOLVED','CLOSED'))              AS Resolved_Complaints,
    COUNT(*)                                                               AS Total_Complaints
FROM Department d
LEFT JOIN Complaint c ON c.Department_ID = d.Department_ID
GROUP BY d.Department_ID, d.Department_Name;

\d+ vw_department_workload
CREATE VIEW
                              View "public.vw_department_workload"
      Column       |          Type          | Collation | Nullable | Default | Storage  | Description
-------------------+------------------------+-----------+----------+---------+----------+-------------
 department_id     | integer                |           |          |         | plain    |
 department_name   | character varying(100) |           |          |         | extended |
 open_complaints   | bigint                 |           |          |         | plain    |
 resolved_complaints | bigint               |           |          |         | plain    |
 total_complaints  | bigint                 |           |          |         | plain    |
View definition:
 SELECT d.department_id,
    d.department_name,
    count(*) FILTER (WHERE c.status::text = ANY (ARRAY['OPEN'::character varying, 'IN_PROGRESS'::character varying, 'ESCALATED'::character varying]::text[])) AS open_complaints,
    count(*) FILTER (WHERE c.status::text = ANY (ARRAY['RESOLVED'::character varying, 'CLOSED'::character varying]::text[])) AS resolved_complaints,
    count(*) AS total_complaints
   FROM department d
     LEFT JOIN complaint c ON c.department_id = d.department_id
  GROUP BY d.department_id, d.department_name;

civic_grievance=#
```

```
civic_grievance=# \dt
             List of relations
 Schema |       Name       | Type  |  Owner
--------+------------------+-------+----------
 public | User             | table | postgres
 public | complaint        | table | postgres
 public | department       | table | postgres
 public | escalation       | table | postgres
 public | notification     | table | postgres
 public | status_history   | table | postgres
 public | upvote           | table | postgres
 public | ward             | table | postgres
(8 rows)

civic_grievance=# \dv
                  List of relations
 Schema |            Name             | Type |  Owner
--------+-----------------------------+------+----------
 public | vw_complaint_full_detail    | view | postgres
 public | vw_department_workload      | view | postgres
 public | vw_escalation_chain         | view | postgres
 public | vw_sla_breach_candidates    | view | postgres
(4 rows)

civic_grievance=#
```